

EDUGENIE

6.2 Accessibility & Universal Design

Inclusive interaction, sensory support, and flexible learning access

Introduction

Accessibility and Universal Design aim to make EDUGENIE usable by learners with different abilities, devices, communication needs, and sensory preferences. An inclusive environment should provide multiple ways to understand information, interact with the system, and demonstrate learning.

Multiple Interaction Methods

- Voice commands for students who have difficulty using conventional controls.
- Keyboard navigation for learners who cannot depend on mouse-based interaction.
- Simple button controls for reduced interaction complexity.
- Alternative input devices where supported by the operating system or hardware.
- Visual assistance such as readable text, clear contrast, labels, and scalable interface elements.

Audio and Sensory Support

- Audio descriptions can explain important visual events.
- Captions or transcripts can provide text alternatives to spoken information.
- Motion reduction can limit unnecessary animation.
- Adjustable sound effects and background audio can reduce sensory overload.
- Students can benefit from simplified interfaces with fewer simultaneous controls.

Universal Design Principles

- Provide more than one way to access information.
- Provide more than one way to interact with learning activities.
- Provide more than one way to demonstrate understanding.
- Keep accessibility settings visible and adjustable.
- Test the interface with real users and revise based on feedback.



Needs → accessible controls → learning environment → response

Comparison / Learning Table

Need	Possible feature	Purpose
Motor access	Keyboard / alternative input	Reduce dependence on mouse
Visual access	Text scaling / audio description	Support information access
Hearing access	Captions / transcripts	Provide text alternatives
Sensory needs	Reduced motion / audio controls	Reduce overload
Cognitive support	Simple controls / guided steps	Reduce interaction complexity

Sample Coding Concept

```
# Simplified accessibility preference model settings = { "captions": True, "audio_description":  
True, "reduced_motion": True, "keyboard_mode": True } if settings["reduced_motion":  
animation_speed = 0 if settings["captions"]: show_captions = True print(settings)
```

Extra Project Content

- Classroom activity: Give students the same interactive task with different interface settings and discuss which features improve usability.
- Accessibility testing: Check keyboard-only navigation, readable text, captions, audio alternatives, focus order, and motion controls.
- Assessment idea: Allow students to submit answers through supported input methods while keeping the academic criteria consistent.
- Future extension: Build an accessibility profile that students can adjust at any time and carry across multiple EDUGENIE lessons.
- Important design principle: Accessibility should be part of the system architecture, not a final add-on.